



Zoo 2025

Digital Securities, Quantum-Secure Settlement,
and the Holographic Consensus DAO

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Abstract

We articulate five intertwined planks that together describe ZOO Labs Foundation as it stands at the QUASAR 3.0 activation date (2025-12-25): (i) the canonical team and mission that descend from the October 31, 2021 founding whitepaper; (ii) ZOO’s positioning against the ~\$113 trillion global digital-securities market through its exclusive partnership with LIQUIDITY and the LUX chain stack; (iii) the quantum-secure settlement guarantee provided by QUASAR 3.0 (LUX LP-020), which composes three independent cryptographic hardness assumptions (BLS₁₂₋₃₈₁, Ringtail Ring-LWE, and ML-DSA-65 via Z-Chain Groth16); (iv) a Homomorphic / Holographic Consensus DAO with weighted voting over *advocacy*, *involvement*, *contribution*, and *token stake*; and (v) democratized end-to-end information access powered by per-LLM chains, public verifiable inference receipts, and zero-knowledge selective disclosure. This paper is the canonical cross-reference for these five planks. Spec freeze: 2025-12-15. Activation alongside QUASAR 3.0: 2025-12-25.

Keywords: digital securities, holographic consensus, FHE voting, QUASAR 3.0, post-quantum settlement, weighted governance, open AI protocol

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1 Team and Mission

1.1 Founding

ZOO Labs Foundation Inc. is a 501(c)(3) non-profit founded by **Antje Worring**, who authored the original Zoo whitepaper on October 31, 2021. That document is the canonical founding artifact of the foundation; it describes a Game-Fi multiverse for endangered species, emotionally-intelligent NFT animals, and a DAO that routes economic value back to wildlife preservation. Worring also coined the term *NFT Liquidity Protocol* in the January 2022 deck, which is the lineage that today underwrites the broader Liquidity Protocol family of on-chain settlement primitives.

1.2 Mission

The ZOO mission, as carried forward to the QUASAR 3.0 activation (2025-12-25), is:

An open AI protocol for research aligned with environment-saving, endangered AI, and the intersection of AI, blockchain, metaverse, and gaming.

This articulation is intentionally broader than “conservation crypto.” ZOO now spans frontier-AI research (the ZEN model family, per-LLM chains, Hamiltonian LLMs, Z-JEPA, BitDelta), metaverse and gaming infrastructure (Game-Fi, virtual habitats, play-to-serve mechanics), conservation and DeSci (federated wildlife monitoring, satellite ecology, citizen science protocols), and governance (the holographic DAO described in §4).

1.3 Core team

- **Antje Worring** — Founder, Zoo Labs Foundation. Author of the original Oct 31, 2021 whitepaper. antje@zoo.ngo.
- **Zach Kelling** — Chief Scientist / engineering lead. Co-author on the Foundation Mission paper, the per-LLM chains paper, and on this document. Drives the ZEN model family, the per-LLM chain integration with LUX QUASAR 3.0, and the Hanzo / Lux / Zoo cross-stack. zach@zoo.ngo.
- **Research staff** — authors and maintainers of the 80+ paper series and the 130+ ZIP corpus (research@zoo.ngo).
- **Engineering staff** — maintainers of `zoo/node` (Rust), `zoo/app` (Tauri + React desktop client), `zoo/zoo` (Next.js front-end), and the Zen LM training stack.

1.4 Allied organizations

ZOO operates inside an explicitly layered ecosystem:

- **LUX Network** provides regulated chain infrastructure and the QUASAR 3.0 post-quantum consensus engine (LUX LP-020, see §3).
- **HANZO AI Inc.** (Techstars ’17) operates the Hanzo AI Chain inference and commerce marketplace and co-develops the ZEN model family.
- **LIQUIDITY** is ZOO’s exclusive partner for retail-scale digital-securities trading; it brings the regulated ATS / broker-dealer / transfer-agent perimeter (see §2).

1.5 Advisors and contributors

The foundation maintains a rolling advisory roster across conservation science, AI research, DeSci / open-research, and governance. The advisor set is governed by the meta-DAO and updated through

the contribution-graph attestation pipeline anchored on LUX A-Chain (LUX LP-134). Detailed roster and join paths are in `zips/team.md` (canonical team page).

1.6 Provenance vs. today

The Oct 31, 2021 founding vision named three pillars: open AI, blockchain, and metaverse. Four years on, those pillars have hardened into specific protocols: per-LLM chains for open AI (one sovereign QUASAR appchain per model, see [10]), LUX QUASAR 3.0 for blockchain, and a metaverse layer that manifests across the Game-Fi, gaming, and AR/VR pillars of the original whitepaper. The remainder of this paper specifies how those pillars now extend into digital securities, quantum-secure settlement, holographic governance, and democratized information access.

2 \$113 Trillion Digital Securities Market — Powered by Lux + Liquidity

2.1 Market size

The total addressable market for digital securities globally is estimated at approximately **\$113 trillion**. This figure is the standard market-size estimate aggregating global capitalization across listed equities, fixed income, derivatives, and tokenized real-world assets [14, 15]. We are explicit that this is a *market-size estimate*, not a ZOO-specific total-value-locked (TVL) claim. The ZOO ecosystem is not attempting to custody \$113 trillion; it is positioning itself as the open AI / wildlife / metaverse layer of an ecosystem whose addressable opportunity is that large.

2.2 Positioning

The stack of responsibilities at the QUASAR 3.0 activation date is:

- **Zoo Labs Foundation** provides the open AI / wildlife / metaverse layer. ZOO is the application surface — per-LLM chains, NFT-native conservation, AR/VR companions, Game-Fi.
- **LUX Network** provides the regulated chain infrastructure: QUASAR 3.0 consensus (LUX LP-020), identity proof on Z-Chain, custody on M-Chain MPC, and FHE substrate on F-Chain. Settlement happens here.
- **LIQUIDITY** is ZOO's *exclusive partner* for actual securities trading at retail scale. LIQUIDITY carries the regulated ATS / broker-dealer / transfer-agent perimeter [13] and is where the regulated market center, control plane, and official records live.

2.3 Tagline and concrete capabilities

Robinhood-style capabilities for everyone — actually make money like the 1% finally.

What that resolves to in concrete economic terms:

Capability	Description
Zero-commission trading	No per-trade commission for retail orders routed through LIQUIDITY. Revenue is earned on spread, market-data, and subscription tiers, not on commission per share.
Fractional shares	Position sizes denominated to the cent, not to the share. A \$5 position in a \$200 stock is one-fortieth of a share.

Capability	Description
Instant settlement on QUASAR	Trades settle in sub-second finality on QUASAR 3.0 consensus (§3). Cash and securities legs both atomic.
Tokenized real-world securities	Stocks, bonds, commodities, REITs, and private-equity positions issued as on-chain tokens with the official transfer-agent record maintained by LIQUIDITY.
24/7 markets	The chain is always on; market hours are an exchange-side policy, not a settlement-layer constraint. Off-hours trading routes via LIQUIDITY as the market center permits.
Public verifiable order book	Public Merkle commitments of every match; every retail trader can verify their fill against the canonical order book without trusting a private feed.
Compliant by construction	KYC, AML, suitability, and best-execution checks are enforced at the LIQUIDITY perimeter <i>before</i> an order is routed to the chain. The chain is not a regulatory loophole; it is the audit-eligible settlement layer behind a regulated front.

2.4 Compliance

The full regulated stack is operated by LIQUIDITY, not by ZOO. LIQUIDITY carries:

- **SEC ATS** registration — the alternative trading system is the regulated market center. ZOO does not operate one.
- **FINRA broker-dealer** registration — order routing, best execution, suitability, KYC/AML.
- **Registered transfer agent** — maintenance of the official record of securities ownership and transfer.

This separation is deliberate: ZOO Labs Foundation is a 501(c)(3) non-profit and cannot itself operate a regulated market center. LIQUIDITY carries that perimeter so ZOO can ship the open-AI / wildlife / metaverse layer that lives on top of it.

2.5 Quantum-secure settlement

Every securities trade routed through this stack settles with a QUASAR 3.0 certificate (see §3): BLS-aggregate classical signature, Ringtail Ring-LWE post-quantum threshold signature, and an ML-DSA-65 identity proof committed via the Z-Chain Groth16 rollup. The retail user can therefore truthfully say:

- *Your trade is post-quantum-safe.*
- *Your custody is post-quantum-safe.*
- *Your audit trail is post-quantum-safe.*

This is the first time those three properties hold simultaneously for a retail-accessible securities stack.

2.6 Why Zoo (and not a generic chain)

ZOO is the alignment layer that gives the stack a reason to exist beyond pure financial throughput. Conservation-aligned issuance, NFT companions that fund real-world sanctuaries, and per-LLM chains that track research provenance are the durable economic flywheels that keep retail traders engaged longer than a fee-rebate program would. The ZOO Foundation’s 501(c)(3) status binds

a portion of every economic flow back into endangered-species preservation and open AI research, which is the lineage of the founding 2021 vision.

3 Quantum-Secure Settlement

3.1 Three independent hardness assumptions

QUASAR 3.0 (LUX LP-020, [1, 2]) provides finality based on *three* independent cryptographic hardness assumptions, composed in a single block-finalization certificate. An adversary must break *all three* simultaneously to forge a LUX/ZOO block.

Layer	Hardness assumption	Role in finality
BLS ₁₂₋₃₈₁	Discrete log over BLS pairings (classical). See LUX LP-075 [7].	Classical aggregate signature for the validator quorum. Carries the bulk of liveness throughput.
Ringtail Ring-LWE	Ring Learning-With-Errors (post-quantum lattice). See LUX LP-073 [6, 18].	Post-quantum threshold signature; co-signed with BLS on every block. Falls back to majority Ringtail-only certificate if BLS is suspect.
ML-DSA-65 via Z-Chain Groth16	Module-Lattice signature (NIST FIPS 204). See LUX LP-070 [5, 17], with succinct proof of validator identity via Groth16 [16].	Post-quantum <i>identity proof</i> layer; binds each block-signing key to a PQ-safe credential committed on Z-Chain.

3.2 Forgery requires breaking all three

A valid QUASAR 3.0 block-finalization certificate σ contains:

$$\sigma = (\sigma_{\text{BLS}}, \sigma_{\text{Ringtail}}, \pi_{\text{ML-DSA}})$$

where σ_{BLS} is the aggregate BLS₁₂₋₃₈₁ signature from the validator quorum, σ_{Ringtail} is the threshold Ring-LWE signature, and $\pi_{\text{ML-DSA}}$ is a Groth16 proof that every signing key was bound to a valid ML-DSA-65 credential at block height h .

- Forging σ_{BLS} alone requires breaking discrete log on BLS12-381. Classical adversary; quantum adversary breaks this in polynomial time post-Q-day.
- Forging σ_{Ringtail} requires solving Ring-LWE for the chosen lattice parameters. Believed secure against both classical and quantum adversaries.
- Forging $\pi_{\text{ML-DSA}}$ requires either breaking ML-DSA-65 (a different module-lattice problem from Ring-LWE) or breaking the Groth16 soundness assumption (knowledge of pairing-based extractor).

The hardness assumptions are deliberately heterogeneous: discrete log over pairing-friendly curves (classical), Ring-LWE (lattice family A), and module lattice signatures with succinct proof (lattice family B with pairing-based zero-knowledge). A break in any single family does not propagate to the others.

3.3 Q-day resilience

Define *Q-day* as the day a cryptographically relevant quantum computer breaks BLS_{12-381} in polynomial wall-clock time.

- *Pre-Q-day*: all three signatures required. BLS carries liveness; PQ layers are belt-and-braces.
- *Post-Q-day*: BLS compromise alone does not break finality. The Ringtail + ML-DSA pair is sufficient for Byzantine-fault-tolerant finalization. The chain enters “PQ-only” mode by social-consensus governance vote (Holographic DAO, §4) until BLS is replaced.
- *No silent break*: an attacker who silently obtains a BLS forgery cannot retroactively rewrite history because the Ringtail + ML-DSA proofs remain on-chain in every prior block.

3.4 Settlement guarantees for Zoo / Liquidity workloads

Every securities trade settled via the LUX chain stack carries a QUASAR 3.0 certificate. Three concrete guarantees follow for a retail user:

1. **Trade integrity is post-quantum-safe.** The trade match is signed under all three layers; a future quantum adversary cannot rewrite a settled trade.
2. **Custody is post-quantum-safe.** M-Chain (LUX LP-019 [4]) holds threshold custody keys using Ringtail PQ threshold signatures.
3. **Audit trail is post-quantum-safe.** The Merkle root of every settled block is anchored under QUASAR 3.0; the auditor needs only the PQ verification key set to verify historical state.

3.5 Cross-references

- LUX LP-020 — QUASAR Consensus 3.0 [1].
- LUX LP-070 — ML-DSA-65 profile [5].
- LUX LP-073 — Ringtail Ring-LWE threshold [6].
- LUX LP-075 — BLS_{12-381} aggregate [7].
- LUX LP-105 — QUASAR 3.0 paper [2].
- LUX LP-019 — M-Chain MPC [4].
- LUX LP-013 — F-Chain FHE [3].
- LUX LP-134 — A/B/M/F chain topology [8].

4 Homomorphic / Holographic Consensus DAO

4.1 Two adjectives, two distinct properties

The ZOO DAO is described as both *homomorphic* and *holographic*. They are independent properties:

- **Homomorphic** means votes are tallied while still encrypted. Tallying happens under fully homomorphic encryption (FHE) on LUX F-Chain (LUX LP-013 [3]). The aggregate result is decryptable; individual ballots are not.
- **Holographic** means every part of the DAO encodes the whole’s intent. Nested sub-DAOs (per-LLM chains, per research vertical, per regional chapter) inherit governance topology from the meta-DAO but can also propose and ratify locally. Each shard is structurally a small copy of the whole.

4.2 Homomorphic vote tallying

A vote is encoded as a ciphertext under an F-Chain FHE public key. The protocol:

1. Each participant submits $\text{Enc}_{\text{pk}}(v_i \cdot w_i)$ where $v_i \in \{-1, 0, +1\}$ is the vote (against / abstain / for) and w_i is the participant’s voting weight (§4.4).
2. F-Chain [3] computes $\sum_i \text{Enc}(v_i w_i)$ homomorphically. The encrypted sum is published on-chain.
3. Decryption of the aggregate is performed only by the M-Chain MPC custodial threshold (LUX LP-019 [4]) after the voting window closes.
4. Individual ballots remain encrypted forever. Aggregate result is published with a Quasar 3.0 finality proof.

This eliminates voter coercion: a participant cannot prove how they voted to a coercer because nobody — not even the M-Chain custodians acting jointly — can decrypt an individual ciphertext. Only the sum is decryptable.

4.3 Holographic structure

The meta-DAO is the apex. Sub-DAOs exist for:

- Each per-LLM chain (one sub-DAO per Zen model; see [10]).
- Each research vertical (Conservation AI, Educational AI, Frontier AI, DeSci).
- Each regional chapter (geographic conservation cells).

Each sub-DAO inherits the meta-DAO’s governance topology (weighted voting, FHE tally, M-Chain custody) and can propose and ratify locally on issues scoped to its shard. Issues that cross shards escalate to the meta-DAO. The structural property is *self-similarity*: a sub-DAO looks like a smaller meta-DAO in every governance dimension.

This is what we mean by *holographic*: every part contains the whole’s pattern. A sub-DAO can fork into its own chain with the same governance machinery and remain interoperable with the meta-DAO without bespoke integration.

4.4 Weighted voting

Voting weight is computed as a weighted sum across four axes:

- **Advocacy** — public posts, citations, and social engagement, attested through Z-Chain (anti-Sybil; see below).
- **Involvement** — contributor activity: commits, code reviews, training submissions, ceremony participation, validator uptime.
- **Contribution** — landmark research artifacts, model releases, novel architectures, dataset curation, reproducibility attestations.
- **Token stake** — locked \$ZOO stake. Always present but deliberately the smallest weight, to prevent plutocracy.

The formula:

$$\text{weight}(p) = \alpha \cdot a_p + \beta \cdot i_p + \gamma \cdot c_p + \delta \cdot s_p \quad (1)$$

subject to $\alpha + \beta + \gamma + \delta = 1$, where a_p, i_p, c_p, s_p are the participant’s normalized advocacy, involvement, contribution, and token-stake scores respectively.

Default initial weights at activation (2025-12-25) are governed by the meta-DAO and start as:

$$(\alpha, \beta, \gamma, \delta) = (0.20, 0.30, 0.40, 0.10)$$

i.e. 20% advocacy, 30% involvement, 40% contribution, 10% token stake. These weights can be adjusted by meta-DAO vote (itself a homomorphic, weighted vote).

4.5 Anti-Sybil

Each input score is gated by attestation:

- Advocacy (a_p) requires the social platform to be attestation-rooted (i.e. the platform itself runs an attestation endpoint anchored on Z-Chain). Posts on un-rooted platforms do not count.
- Involvement (i_p) is computed from the contribution-graph attestation pipeline anchored on LUX A-Chain (LUX LP-134 [8]). Each commit and review is signed by a TEE-attested identity.
- Contribution (c_p) requires reproducibility attestation (per ZIP-0606 in the Zoo ZIP corpus); a paper, model, or dataset must verify before it confers contribution weight.
- Token stake (s_p) is unforgeable on-chain by definition but is rate-limited by the $\delta = 0.10$ default to bound plutocracy.

A purely token-rich attacker can buy at most $\delta = 0.10$ of total voting weight. To dominate, the attacker must also generate provable advocacy, involvement, and contribution — which the attestation pipeline rejects unless backed by real activity.

4.6 Threshold custody of governance keys

Governance keys (the master key for upgrade proposals, treasury movements above policy thresholds, and emergency halts) are held in threshold custody on M-Chain (LUX LP-019 [4]). A successful proposal pulls a threshold partial signature from each custodian; the resulting Ringtail PQ aggregate signature authorizes the on-chain action. No single custodian can act unilaterally; no custodian can be compelled by physical coercion of one party.

4.7 Cross-references

- LUX LP-013 — F-Chain FHE substrate (vote tally) [3].
- LUX LP-019 — M-Chain MPC (governance key custody) [4].
- LUX LP-134 — A/B/M/F topology (contributor attestation on A-Chain) [8].
- ZIP-0017 — DAO Governance Framework (Zoo ZIP corpus).
- ZIP-0026 — Ecosystem Reputation System (the involvement-score input).

5 Democratized End-to-End Access to Information

5.1 Mission statement

ZOO treats access to information as a first-class protocol primitive, not a UX afterthought. The mission, in one sentence:

Every step from user to model to data to output is verifiable, attributable, and rewards flow to contributors automatically.

End-to-end is the load-bearing word: the property must hold from the user’s keystroke through the model’s inference, through the training data lineage, all the way to the output, and back through the reward distribution. Skipping any step undermines the chain of custody.

5.2 Open AI protocol

ZOO per-LLM chains [10] commit:

- **Model weights** — every checkpoint root is Merkle-committed on the per-LLM chain. Weights are open-weight (downloadable); the chain commits the hash, not the bytes.

- **Training-data lineage** — every training shard’s hash is on the receipts tree. Anyone can audit which examples a model saw.
- **Research artifacts** — every architecture decision, benchmark result, and ablation is registered as an on-chain artifact.
- **Contribution graph** — attribution from output back to the people whose work fed the model is canonical and on-chain.

Compared to closed AI labs (OpenAI, Anthropic), this is a different ontological commitment. Closed labs treat the weights, the data, and the architecture as competitive secrets. ZOO treats them as *public goods* whose provenance is the protocol.

5.3 Public verifiable inference receipts

Every inference call to a Hanzo AI Chain endpoint produces a receipt signed by the TEE-attested inference node and committed to the A-Chain attestation ledger [8, 9]. The receipt contains:

- Hash of the request prompt (or its commitment if private).
- Hash of the response.
- Model identifier and chain block height of the model’s active checkpoint.
- Wall-clock latency, token count, and compute cost.
- TEE attestation that the response was generated by the named model on attested hardware.

A user who pays for an inference call has cryptographic proof that the response was generated by the model they paid for, not a cheaper substitute. A regulator who audits a model’s outputs has a tamper-proof log of every external use.

5.4 Zero-knowledge selective disclosure

The user-side counterpart of public receipts is *selective disclosure*. A user proves eligibility or capability without revealing identity:

- “*I am over 18*” without revealing date of birth.
- “*I hold an accredited-investor attestation*” without revealing net worth.
- “*I am a verified researcher with publication credit on topic X*” without revealing institutional affiliation.
- “*I have voted in this DAO before*” without revealing past votes (consistent with the homomorphic vote tally, §4).

These proofs are issued as Groth16 attestations [16] anchored on Z-Chain and verified on whichever LUX sub-chain needs the gating decision. The verifier learns only the predicate; nothing else leaks.

5.5 Open research registry

Every ZEN model release, every benchmark run, every architecture decision made by the foundation is committed as an on-chain research artifact. This includes negative results (“ablation X did not improve perplexity”), which centralized labs typically suppress.

The registry is queryable: a researcher can ask, “which architecture choices have been tried for problem Y?” and receive a canonical answer drawn from the on-chain artifacts of every per-LLM chain. This is what we mean by an *open research registry* as opposed to scattered preprints.

5.6 Contrast with closed labs

Property	Closed labs	Zoo open AI protocol
Model weights	Proprietary, undisclosed	Open-weight; hash committed on-chain
Training data	Undisclosed; sometimes denied	Lineage hash on per-LLM receipts tree
Inference proof	API trust	TEE-attested receipt on A-Chain
Architecture provenance	Marketing post	On-chain artifact registry
Reward to contributors	None	Attribution-graph payments per per-LLM chain
Audit by regulator	On request, contractually	On-chain by default

5.7 Democratized end-to-end means

Putting it together: *democratized end-to-end access to information* in the ZOO stack means:

1. **User** — can prove eligibility without revealing identity (zero-knowledge selective disclosure).
2. **Model** — weights, architecture, and training data are on-chain provenanced and open-weight.
3. **Data** — every training shard’s lineage is auditable; every contributor receives attribution.
4. **Output** — every inference call produces a verifiable receipt; every output can be traced back to the model checkpoint and ultimately to the contributors who shaped it.
5. **Reward** — royalties and attribution payments flow back along the contribution graph automatically, without a central treasurer.

This is the operational meaning of ZOO’s mission to be “an open AI protocol for research aligned with environment-saving, endangered AI, and the intersection of AI, blockchain, metaverse, and gaming.”

5.8 Cross-references

- [10] — per-LLM chains paper.
- [9] — Hanzo AI Chain whitepaper.
- LUX LP-134 — A-Chain attestation topology [8].
- ZIP-0405 — conservation-aware language models.
- ZIP-0407 — decentralized AI training architecture.
- ZIP-0413 — foundation language model architecture.
- ZIP-0606 — reproducibility attestation.

6 Conclusion

The five planks documented above are not separate initiatives. They are five faces of a single design.

- The **team and mission** (§1) state who is responsible and what the foundation is for.
- The **\$113 trillion digital securities** positioning (§2) is what the ZOO + LUX + LIQUIDITY stack is *addressing* commercially.
- **Quantum-secure settlement** (§3) is the cryptographic floor under everything else: the reason the securities-trading retail user, the DAO voter, and the open-AI researcher can all rely on the same chain.
- The **Homomorphic / Holographic Consensus DAO** (§4) is how decisions get made without coercion, capture, or plutocracy.

- **Democratized end-to-end access to information** (§5) is the alignment objective: what the protocol is optimizing for in the long run.

The foundation activates these planks alongside LUX QUASAR 3.0 on **2025-12-25**. The spec freeze date for this document is **2025-12-15**.

In the spirit of the original 2021 founding vision: every animal raised, every NFT minted, every conversation with an avatar, every trade settled, every model trained, and every vote tallied contributes to the survival of a real species, the integrity of the open AI protocol, and the durability of the ZOO ecosystem.

— *Antje Worrying & Zach Kelling, Zoo Labs Foundation, 2025-12-15.*

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